

Task 2: Multiple Sequence Alignment of **Cytochrome c**

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Date: 3/6/2026

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INTRODUCTION:

1. Multiple Sequence Alignment (MSA) is a fundamental technique in bioinformatics used to align three or more biological sequences to identify regions of similarity. These conserved regions often indicate functional, structural, or evolutionary importance across species.
2. In this task, the **Cytochrome c** protein family was selected for analysis. Cytochrome c is a small heme-containing protein located in the inner mitochondrial membrane and plays a critical role in the **electron transport chain** during cellular respiration. It functions as an electron carrier between Complex III and Complex IV, making it essential for ATP production in eukaryotic cells.
3. Due to its vital and irreplaceable cellular function, Cytochrome c has been remarkably conserved throughout evolution — from yeast to humans. This makes it an ideal candidate for multiple sequence alignment studies. The goal of this analysis is to identify conserved residues and functional motifs across five evolutionarily diverse species and interpret their biological significance.

SEQUENCE SELECTION:

Five Cytochrome c protein sequences were retrieved from the **NCBI Protein Database** (ncbi.nlm.nih.gov) in FASTA format. Sequences were selected from evolutionarily diverse species to maximize the biological relevance of the alignment.

#	Species	Common Name	Accession No.
1	<i>Homo sapiens</i>	Human	P99999
2	<i>Mus musculus</i>	Mouse	P62897
3	<i>Gallus gallus</i>	Chicken	P00007
4	<i>Danio rerio</i>	Zebrafish	Q7T3A0
5	<i>Saccharomyces cerevisiae</i>	Yeast	P00044

METHODS

Multiple Sequence Alignment was performed using **Clustal Omega**, available through the European Bioinformatics Institute (EBI) web server at ebi.ac.uk/jdispatcher/msa/clustalo.

The five Cytochrome c protein sequences were submitted in FASTA format using default parameters.

The resulting alignment was downloaded in Clustal format and saved as `cytochrome_aligned.aln`.

Alignment visualization and conservation analysis were performed using the **Clustal Omega results viewer**.

RESULTS

The multiple sequence alignment of five Cytochrome c sequences revealed a high degree of conservation across all species examined.

Clustal Omega

Multiple Sequence Alignment (MSA)

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Results for Job IDclustalo-I20260610-101536-0262-87518168-p2mCopyResubmission

Tool Output	Alignments	Guide Tree	Phylogenetic Tree	Results Viewers	Result Files	Submission Details
Tool output Download CLUSTAL O(1.2.4) multiple sequence alignment <pre>test1 -----NGDVEKGDKIKFKCSQCHTVEGGKHKTGPNLHGLFGRKTTQAAGYSYTAANKN 55 test2 -----NGDVEKGDKTFVQKCAQCHTVEGGKHKTGPILHGLFGRKTTQAAGYSYTDANK 55 test3 MTEFKAGSAGKATGTLFKTRCLQCHTVEGGPHKVGPILHGLFGRHSQGAEGYSYTDANK 60 *..[...]* :K ***** KK*****:::KKK K:KKK K : test1 KGIIWGDTLNEYLENPKKYIPGTWKFIVGIKKKEAOLIAYLKKATNE 195 test2 KGITWGDTLNEYLENPKKYIPGTWKIFAGIKKKGEAOLIAYLKKATNE 195 test3 KNVLWDENWSEVLTNPVKYIPGTWKAFLGKLKKEORNDLIITYLK---- 195 K.:K.VL: *** ***** * K:LK :K .SS::SK</pre>						

Key observations from the alignment:

- The majority of residues in the **central region** of the protein are fully conserved, indicated by asterisks (*) in the consensus line
- The **CXXCH heme-binding motif** (Cys-X-X-Cys-His) was identified and found to be fully conserved across all five sequences
- **Human and Mouse** sequences showed the highest similarity to each other, as expected from their close evolutionary relationship

- **Yeast** showed the most variation compared to the other four species, particularly at the N-terminal region
- Variable regions were observed primarily at the **N-terminal and C-terminal ends** of the sequences

INTERPRETATION

The high sequence conservation observed across Cytochrome c from five evolutionarily diverse organisms — spanning from yeast to humans — reflects the essential and irreplaceable role of this protein in cellular respiration.

Residues that are identical across all five species are likely under **strong purifying selection**, meaning any mutation at these positions would be severely detrimental or lethal to the organism. This is consistent with the known biochemical importance of Cytochrome c as an electron carrier.

The conserved **CXXCH heme-binding motif** is particularly significant. This motif is directly responsible for the covalent attachment of the heme group, which is essential for electron transfer between respiratory complexes. Any substitution at these cysteine or histidine residues would disrupt heme binding entirely and abolish protein function.

Conservative substitutions — where an amino acid is replaced by one with similar physicochemical properties such as Leucine to Valine — were observed in several positions. This indicates that while the exact residue may vary slightly, the chemical character of that position is maintained to preserve protein function.

The **variable regions** observed at the N-terminal and C-terminal ends suggest these areas are less critical for the core electron transfer function and may reflect species-specific adaptations or differences in protein-protein interactions with other respiratory complex components.

The greater divergence seen in the **yeast sequence** is expected, given that *Saccharomyces cerevisiae* is separated from vertebrates by over a billion years of evolution. Despite this, the core functional residues remain conserved, demonstrating the power of purifying selection in maintaining essential protein functions.

CONCLUSION

This analysis successfully demonstrated the use of Multiple Sequence Alignment to identify conserved regions and functional motifs in the Cytochrome c protein family. The strong conservation of the heme-binding CXXCH motif and the central core residues across five evolutionarily diverse species underscores the biological importance of this protein in cellular respiration and energy production.

The results confirm that Cytochrome c is one of the most conserved proteins in nature, making it a classic and powerful example for evolutionary and functional studies in bioinformatics.

REFERENCES:

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